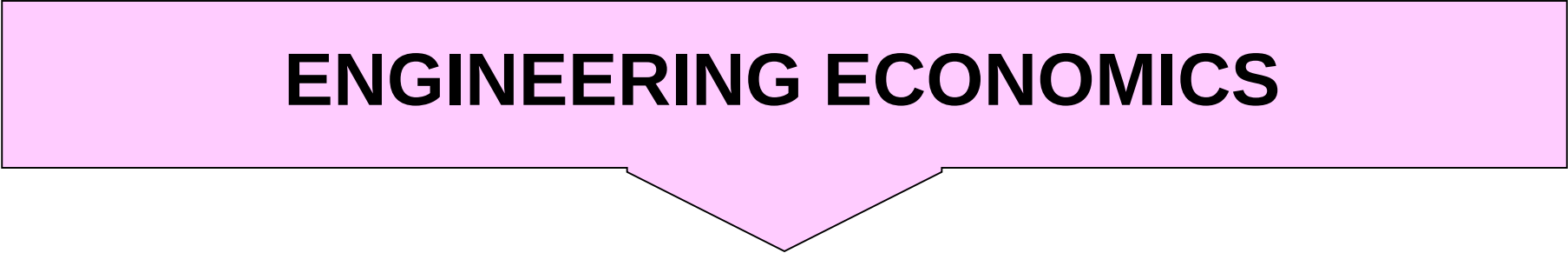


ENGINEERING ECONOMICS



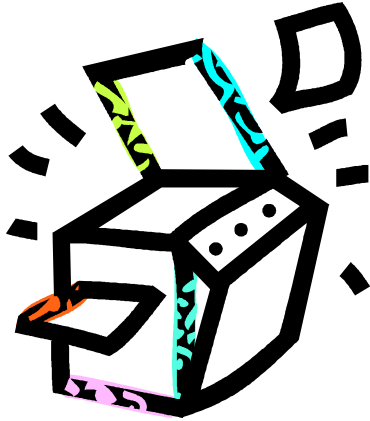
LECTURE - 15



What is Accounting?

Accounting is the recording of all business transactions to provide a financial picture of an organization

Different Accounts



I am a copier. My monetary value to the organization might be found in an account called "Copiers"

I am cash. My value might be found in an account called "Cash"



I am a bill for rent. The amount the organization owes might be found in an account called "Rent Expense"

Note: The amount in a particular account can increase or decrease, depending on the business transaction that affected it

Types of Accounts

- Assets: things of value an organization owns
- Liabilities: obligations an organization owes to someone else
- Expenses: the cost of doing business
- Revenue: income an organization has earned
- Owners' Equity: owners' interest in the company

At least **two** of these account types are involved in any transaction! They can be two of the same or two different account types

Debits and Credits

Account Title Goes Here

Debits:

- ✓ Usually mentioned first
- ✓ Always on the left
- ✓ Must equal credits

Credits:

- ✓ Always on the right
- ✓ Must equal debits

"T-Account"

Debits and Credits

- An **Account** shows the effect of transactions on a given asset, liability, equity, revenue, or expense account.
- **Double-entry** accounting system (two-sided effect).
- Recording done by debiting at least one account and crediting another.
- **DEBITS must equal CREDITS.**

Debits and Credits

Account



- An arrangement that shows the effect of transactions on an account.
- Debit = "Left"
- Credit = "Right"

An Account can be illustrated in a T-Account form.



Account Name	
Debit / Dr.	Credit / Cr.

Debits and Credits

If Debit entries are **greater than** Credit entries, the account will have a debit balance.

Account Name		
	Debit / Dr.	Credit / Cr.
Transaction #1	\$10,000	\$3,000
Transaction #3	8,000	
Balance	\$15,000	

Transaction #2

Debits and Credits

If Credit entries are **greater than** Debit entries, the account will have a credit balance.

Account Name		
Debit / Dr.	Credit / Cr.	
Transaction #1	\$10,000	Transaction #2
	\$3,000	Transaction #3
	8,000	
Balance	\$1,000	

Trial Balance

Trial Balance - a list of each account and its balance; used to prove equality of debit and credit balances.

Acct. No.	Account	Debit	Credit
100	Cash	\$ 140,000	
105	Accounts receivable	35,000	
110	Inventory	30,000	
130	Building	150,000	
200	Accounts payable		\$ 60,000
220	Note payable		150,000
300	Common stock		100,000
330	Retained earnings		
400	Sales		75,000
500	Cost of goods sold	30,000	
		<u>\$ 385,000</u>	<u>\$ 385,000</u>

Adjusted Trial Balance Shows the balance of all accounts, after adjusting entries, at the end of the accounting period.

Preparing Financial Statements

Financial Statements are prepared directly from the Adjusted Trial Balance.

```
graph TD; A[Financial Statements are prepared directly from the Adjusted Trial Balance.] --> B[Balance Sheet]; A --> C[Income Statement];
```

Balance Sheet

Income Statement

Preparing Financial Statements

Assume the following
Adjusted Trial Balance

Adjusted Trial Balance	Debit	Credit
Cash	\$ 140,000	→
Accounts receivable	35,000	→
Building	190,000	→
Note payable		\$ 150,000 →
Common stock		100,000 →
Retained earnings		38,000
Dividends declared	10,000	
Sales		185,000
Interest income		17,000
Cost of goods sold	47,000	
Salary expense	25,000	
Depreciation expense	43,000	
	<u>\$ 490,000</u>	<u>\$ 490,000</u>

Balance Sheet

Balance Sheet

Assets

Cash	\$ 140,000
Accounts receivable	35,000
Building	190,000
Total assets	<u>\$ 365,000</u>

Liabilities

Note payable	150,000
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Stockholders' equity

Common stock	100,000
Retained earnings	115,000
Total liab. & equity	<u>\$ 365,000</u>

Preparing Financial Statements

Assume the following
Adjusted Trial Balance

Adjusted Trial Balance	Debit	Credit
Cash	\$ 140,000	
Accounts receivable	35,000	
Building	190,000	
Note payable		\$ 150,000
Common stock		100,000
Retained earnings		38,000
Dividends declared	10,000	
Sales		185,000
Interest income		17,000
Cost of goods sold	47,000	
Salary expense	25,000	
Depreciation expense	43,000	
	<u>\$ 490,000</u>	<u>\$ 490,000</u>

Income Statement

Income Statement

Revenues:

Sales	\$ 185,000
Interest income	17,000
Total revenue	<u>202,000</u>

Expenses:

Cost of goods sold	47,000
Salary expense	25,000
Depreciation expense	43,000
Total expenses	<u>115,000</u>
Net income	<u>\$ 87,000</u>

Gross income, net income

See Accounting Basic ppt for detail.

Definition of Terms

- **Retained earnings** are the sum of a company's profits, after dividend payments, since the company's inception.
- **Accounts payable (A/P)** are amounts owed to suppliers and other creditors for goods and services bought on credit.
- **Accounts receivable (A/R)** are amounts owed by customers for goods and services a company allowed the customer to purchase on credit.

- **Common stock** represents ownership interests in corporations.
- Shares of common stock provide evidence of ownership in a corporation.
- **Notes payable**

Written promises to pay stated sums of money at future dates.

Discussion